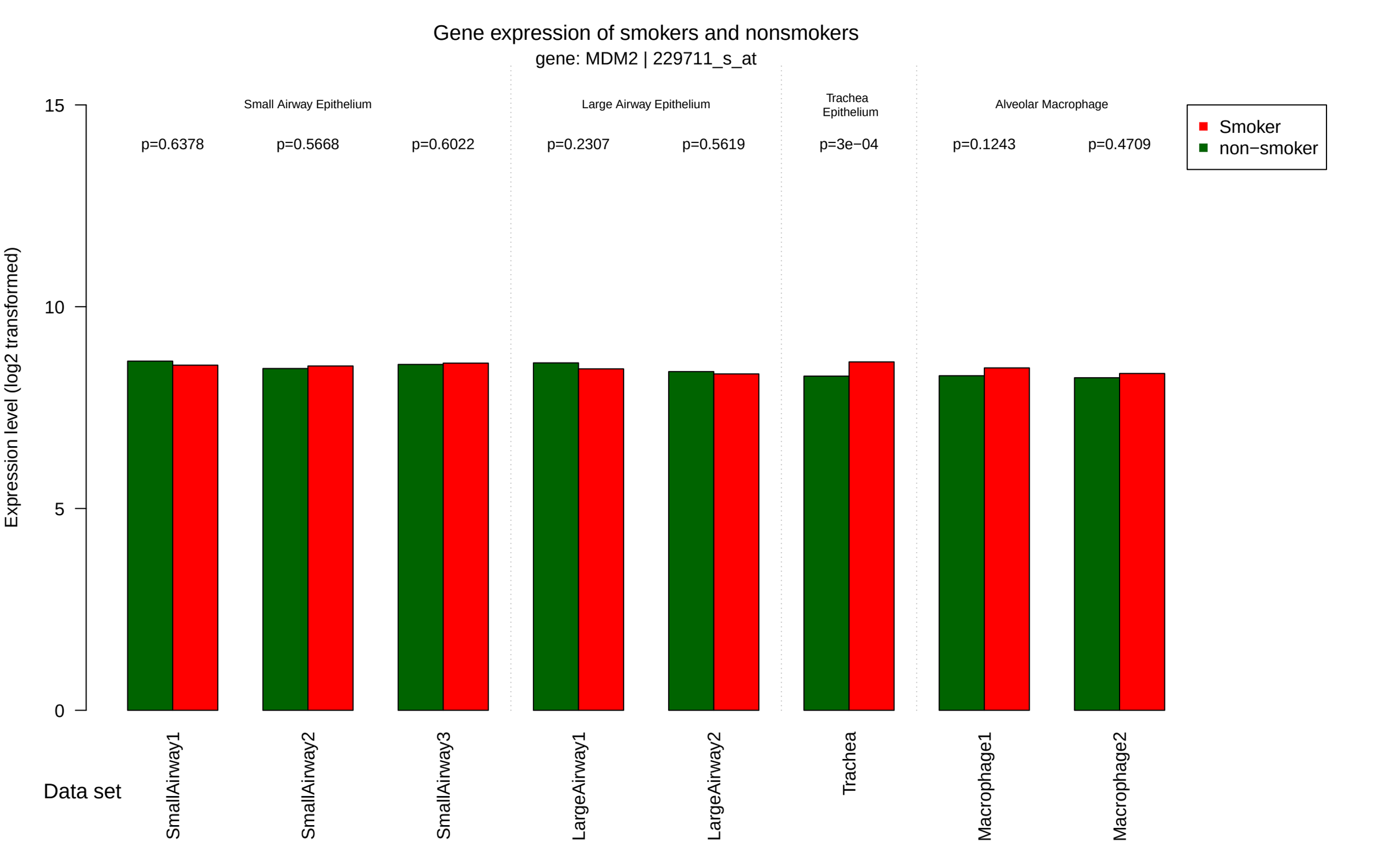
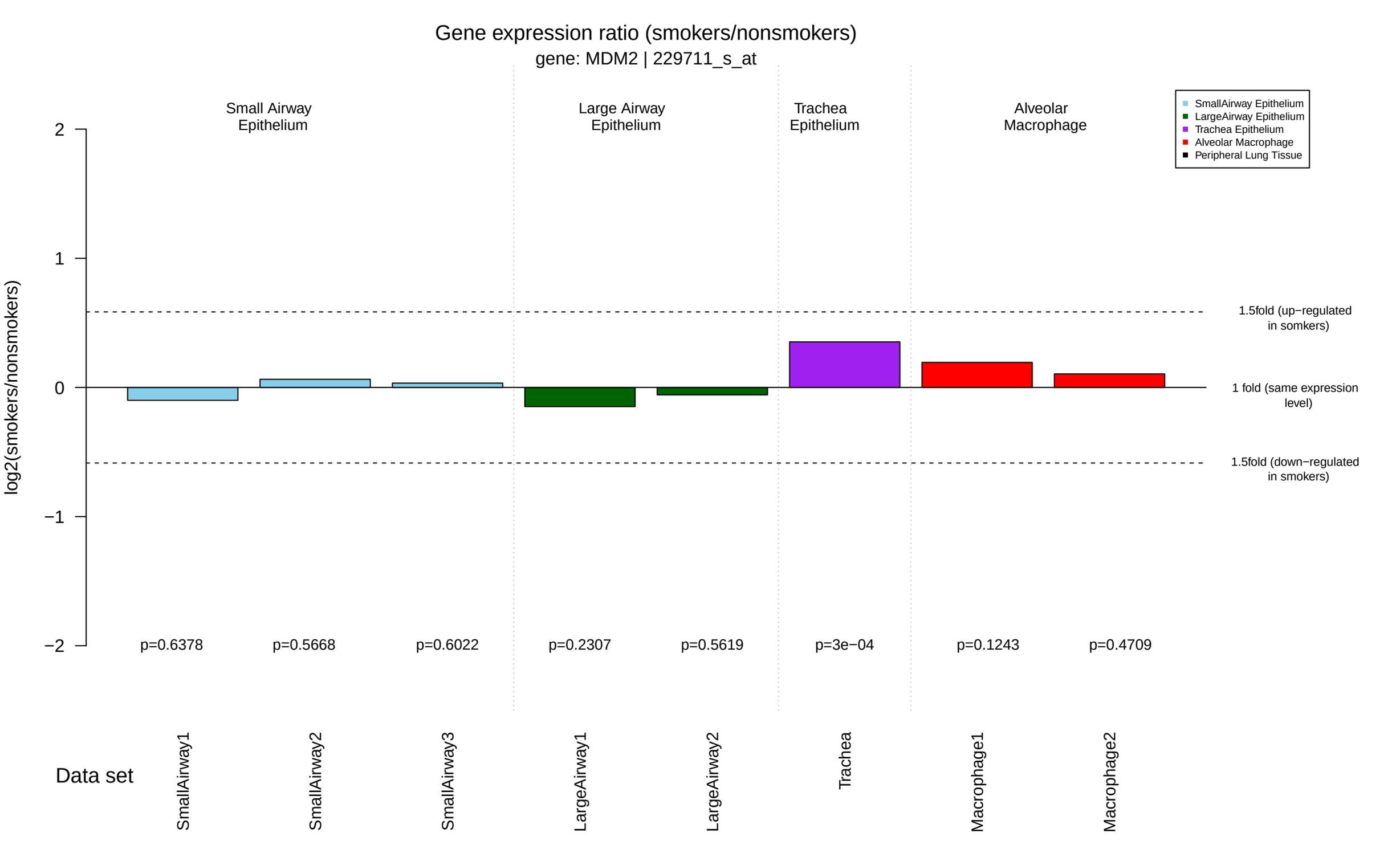
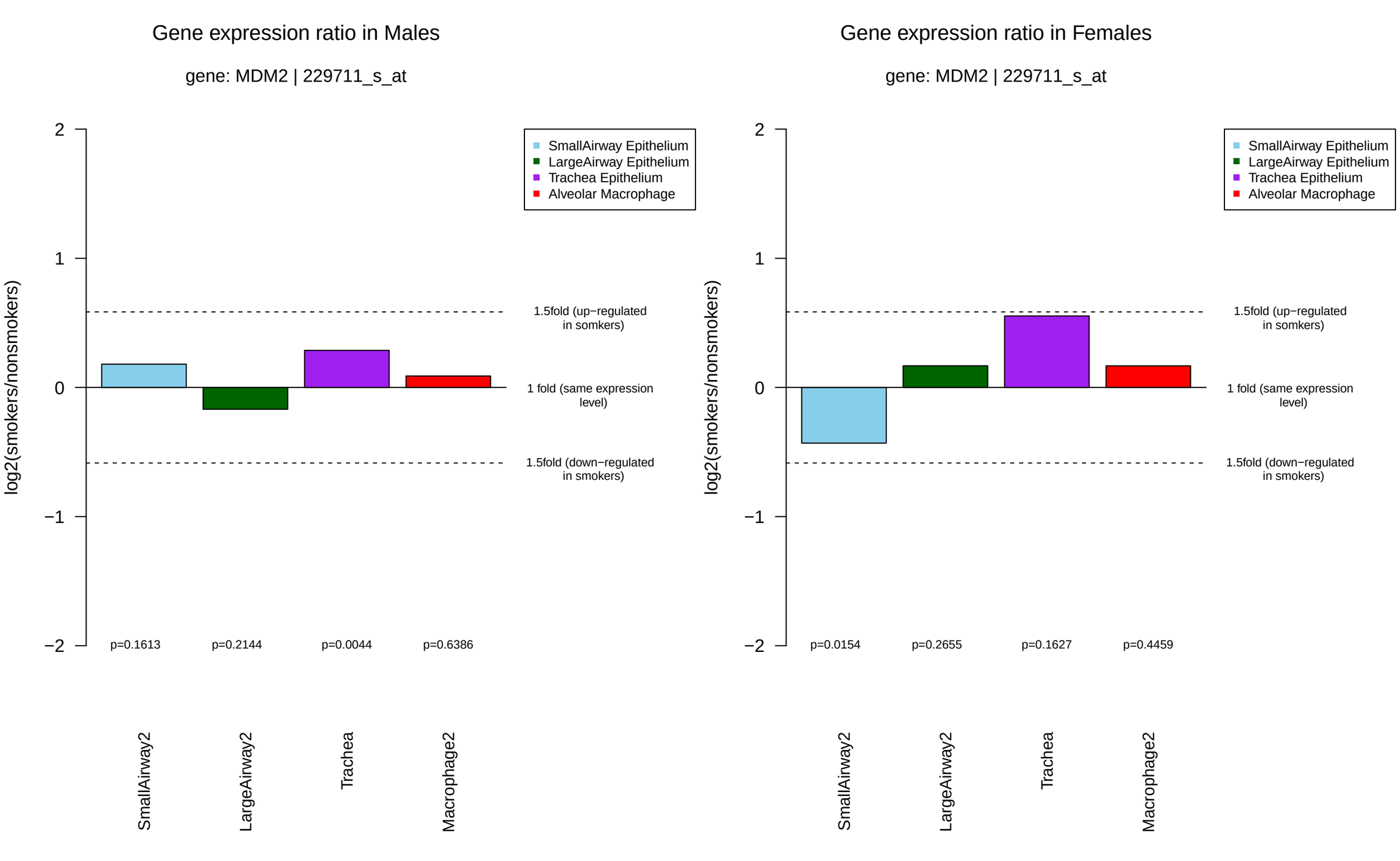
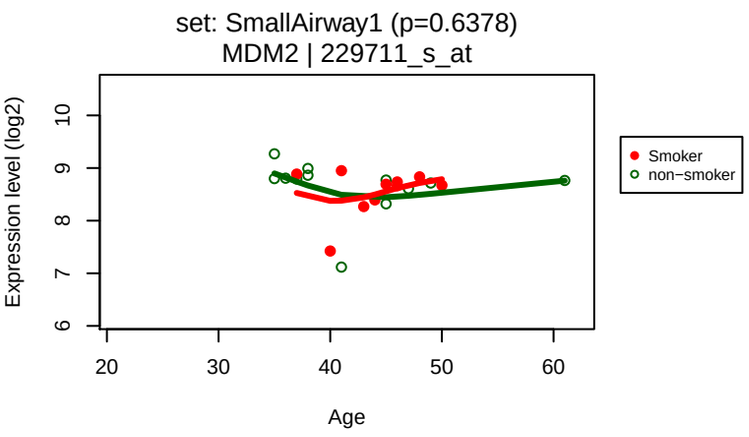
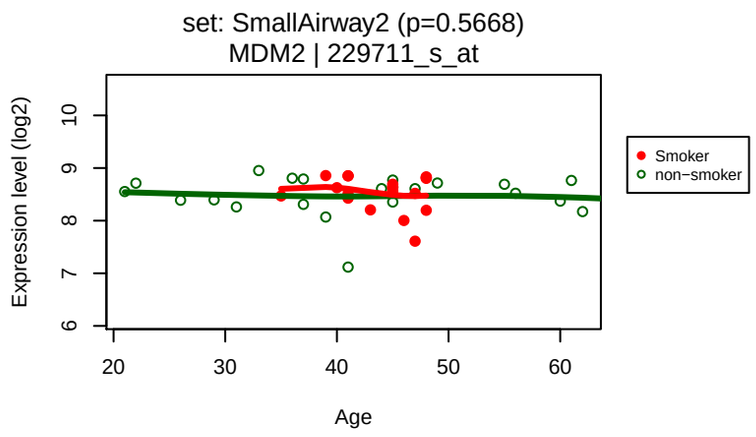
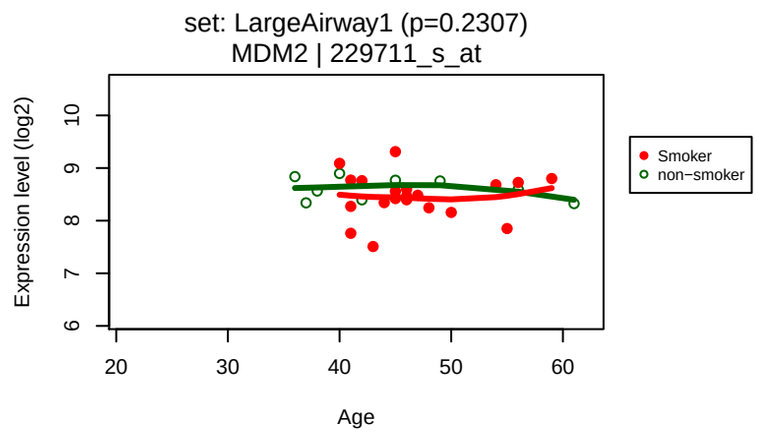

Gene expression of smokers and nonsmokers
gene: MDM2 | 229711_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6378
p=0.5668
p=0.6022
p=0.2307
p=0.5619
p=3e−04
p=0.1243
p=0.4709
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

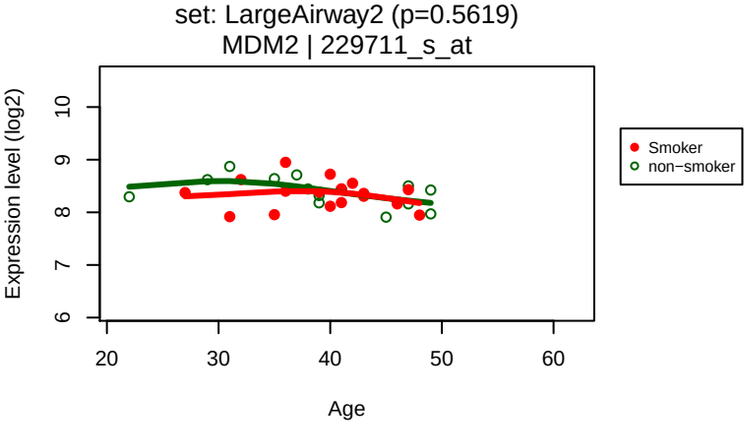

Gene expression ratio (smokers/nonsmokers)
gene: MDM2 | 229711_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6378
p=0.5668
p=0.6022
p=0.2307
p=0.5619
p=3e−04
p=0.1243
p=0.4709
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

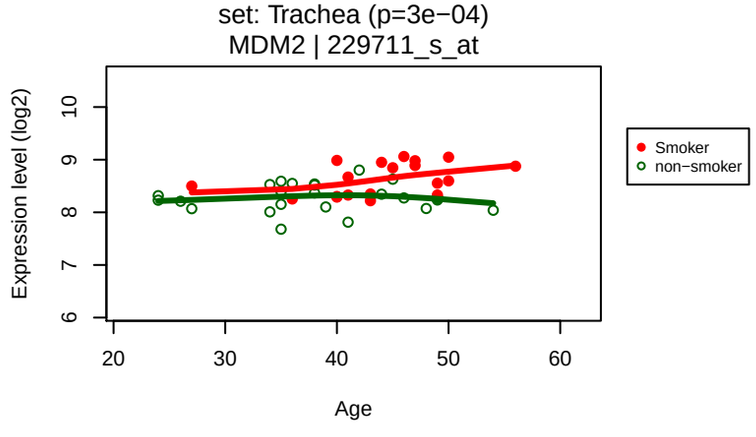

Gene expression ratio in Males
gene: MDM2 | 229711_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1613
p=0.2144
p=0.0044
p=0.6386
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MDM2 | 229711_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0154
p=0.2655
p=0.1627
p=0.4459
SmallAirway2
LargeAirway2
Trachea
Macrophage2

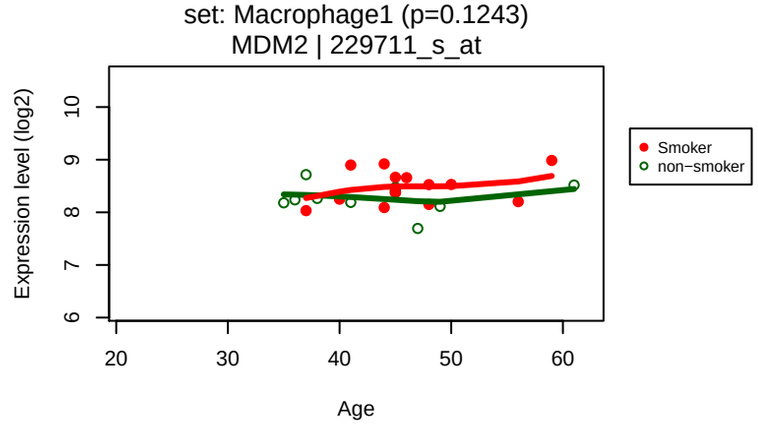

set: SmallAirway1 (p=0.6378)
MDM2 | 229711_s_at
Expression level (log2)
Age
Smoker
non−smoker

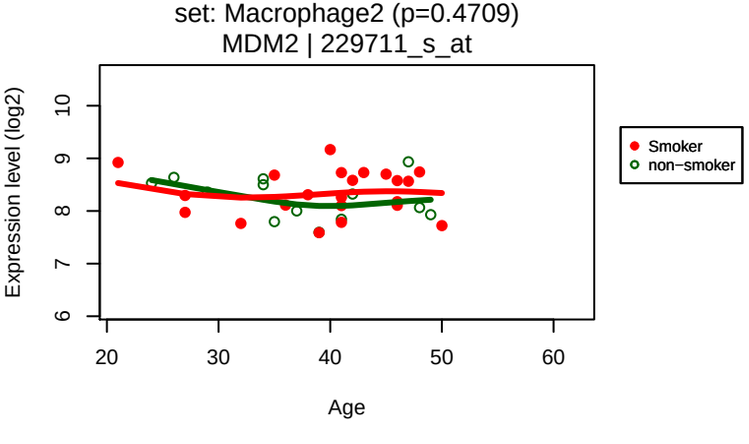

set: SmallAirway2 (p=0.5668)
MDM2 | 229711_s_at
Expression level (log2)
Age
Smoker
non−smoker

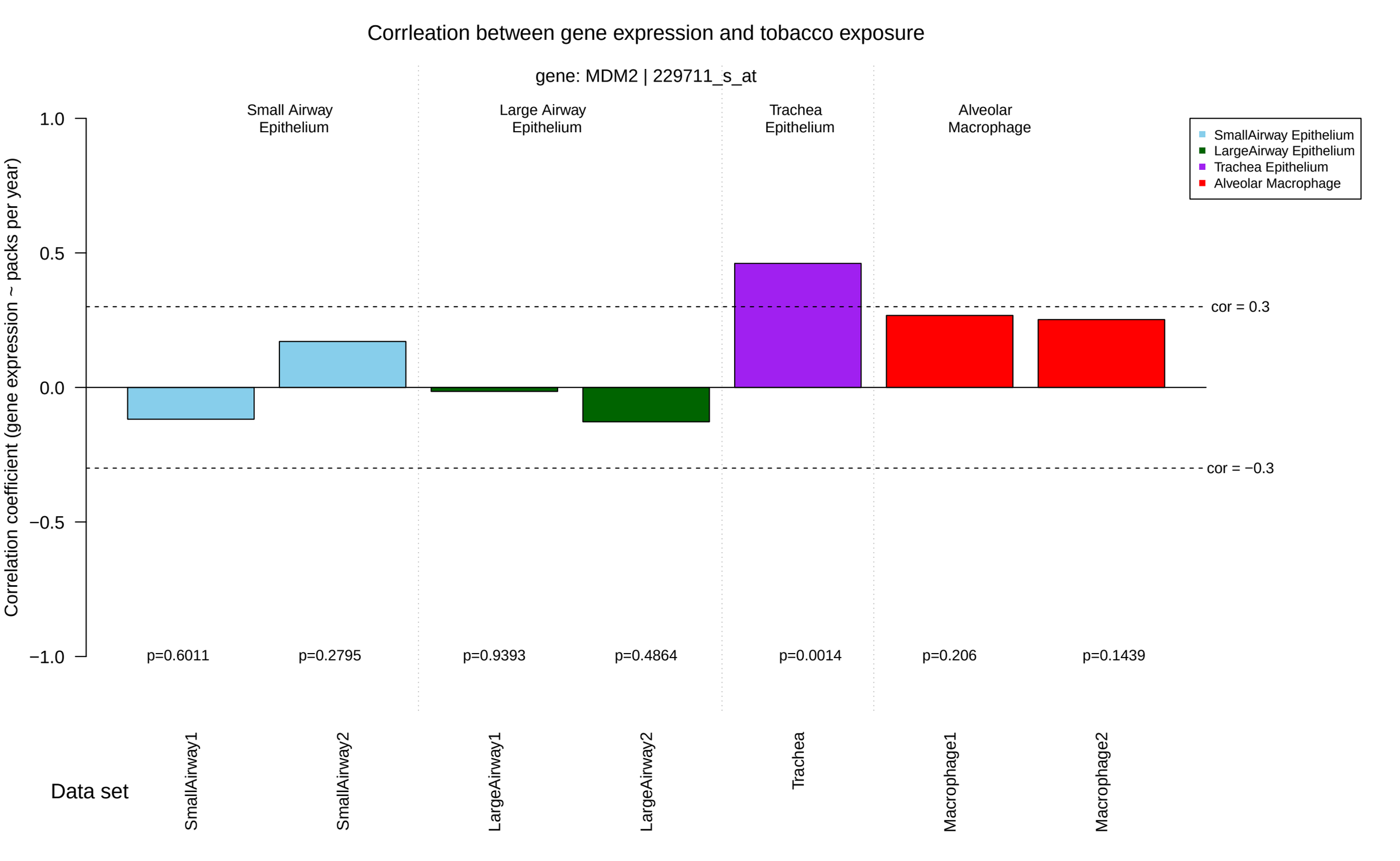

set: LargeAirway1 (p=0.2307)
MDM2 | 229711_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5619)
MDM2 | 229711_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=3e−04)
MDM2 | 229711_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1243)
MDM2 | 229711_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4709)
MDM2 | 229711_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MDM2 | 229711_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6011
p=0.2795
p=0.9393
p=0.4864
p=0.0014
p=0.206
p=0.1439
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2